On Krylov subspace approximations to the matrix exponential operator

Purple team

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Introduction

- Approximation of $exp(\tau A)v$ using Krylov methods, A big matrix, τ step size
- Error bounds for Arnoldi and Lanczos algorithms
- Faster convergence than $(I - \tau A)^{-1}v$ for stiff ODE problems, new integration methods
- Bound $\frac{||\tau A||^m}{m!}$, superlinear convergence for $m \gg ||\tau A||$
- Better behaviour for matrix classes: HND, skew-Hermitian, dependent on eigenvalues distribution

Matrix function definition

- $AV_m = V_m H_m + h_{m+1,m} v_m e_m^T$, $H_m = V_m^* A V_m$
- $q_{m-1}(A)v = V_m q_{m-1}(H_m) e_1 \implies (\lambda I - A)^{-1} v \approx V_m (\lambda I - H_m)^{-1} e_1$
- Numerical range $\mathcal{F}(A) = \{x^* A x : x \in \mathbb{C}^N, ||x|| = 1\}$, $\mathcal{F}(H_m) \subset \mathcal{F}(A)$
- $f(A)v = \frac{1}{i2\pi} \int_\Gamma f(\lambda) V_m (\lambda I - H_m)^{-1} e_1 d\lambda$, Γ is contour around $\mathcal{F}(A)$
- $f(A)v \approx V_m f(H_m) e_1$

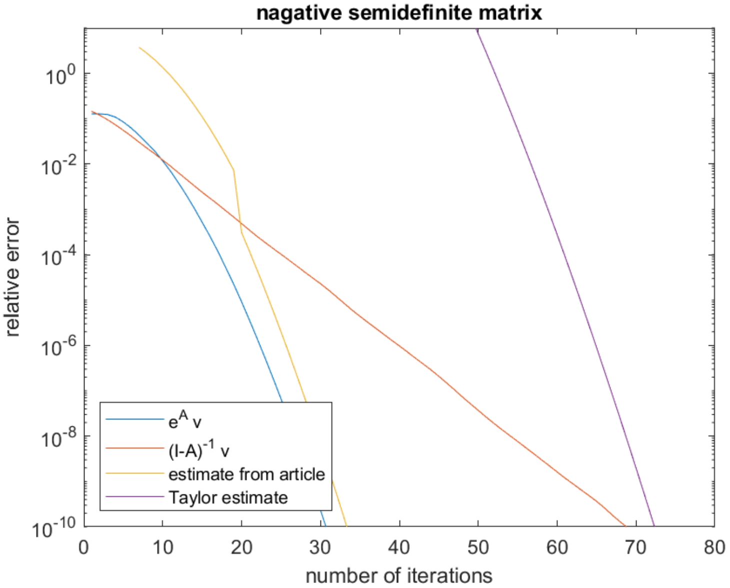
Error bound lemma

- $||f(A)v - V_m f(H_m)e_1|| \leq \frac{M}{2\pi} \int_\Gamma |f(\lambda) - q_{m-1}(\lambda)||\phi(\lambda)|^{-m}|d\lambda|$
- ϕ conformal mapping from exterior of E to exterior of unit circle
- Γ piecewise smooth, f analytic and continuous to boundary
- $M = l(\partial E)/[d(\partial E)d(\Gamma)]$, $l(\partial E)$ length of boundary, $d(S)$ distance between $\mathcal{F}(A)$ and subset S, E convex closed bounded set in complex plane
- Proof idea
- $||(\lambda I - A)^{-1}v - V_m(\lambda I - H_m)^{-1}e_1|| \leq c||p_m(A)||, p_m(\lambda) = 1$
- $p_m(A) = \frac{1}{2\pi i} \int_{\partial E} p_m(z)(zI - A)^{-1}dz \implies ||p_m(A)|| \leq c \max_{z\in E} |p_m(z)|$
- $\max_{z\in E} |p_m(z)| \leq c|\phi(\lambda)|^{-m}$

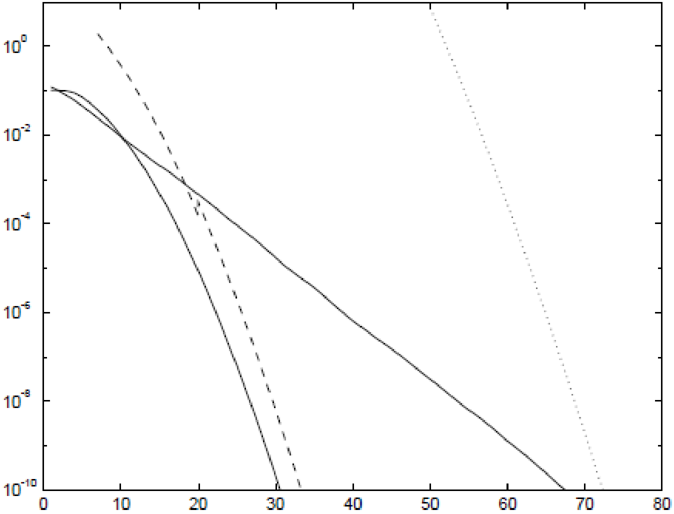
Error bound for Hermitian negative semidefinite matrix

- Eigenvalues in $[-4\rho, 0], \varepsilon_m = || \exp(\tau A)v - V_m \exp(\tau H_m)e_1||$
- $\varepsilon_m \leq 10e^{-m^2/(5\rho\tau)}$ for $\sqrt{4\rho\tau} \leq m \leq 2\rho\tau$
- $\varepsilon_m \leq 10(\rho\tau)^{-1}e^{-\rho\tau}\left(\frac{e\rho\tau}{m}\right)^m$ for $m \geq 2\rho\tau$
- Example: A diagonal with eigenvalues in $[-40, 0]$, v random vector of dimension 1001
- Figure: Convergence compared with linear decay for CG $(I - A)^{-1}v$ and theoretical bound (theorem and Taylor remainder)

Error bound for negative semidefinite matrix



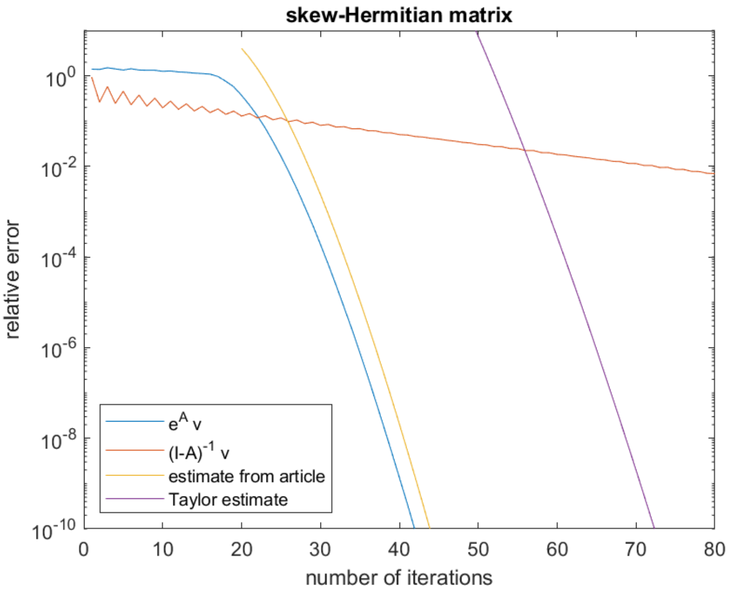
Error bound for negative semidefinite matrix source



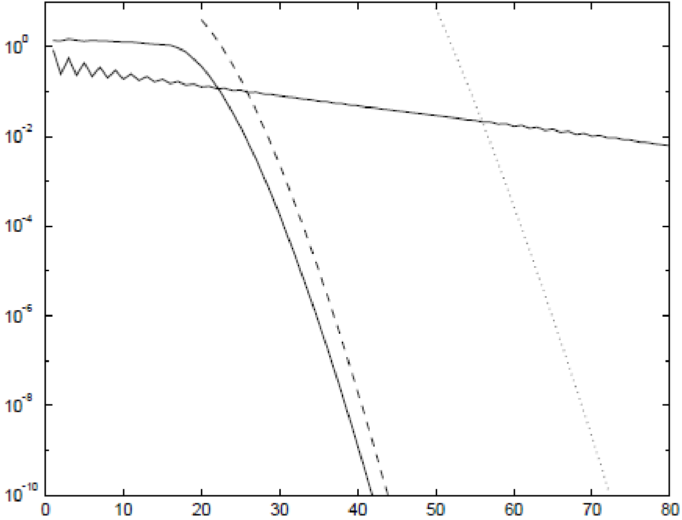
Error bound for skew-Hermitian matrix

- Eigenvalues on imaginary axis in interval of length 4ρ
- $\varepsilon_m \leq 12e^{-(\rho\tau)^2/m}\left(\frac{e\rho\tau}{m}\right)^m$ for $m \geq 2\rho\tau$
- Example: A diagonal with eigenvalues in $[-20i, 20i]$, v random vector of dimension 1001
- Figure: Convergence compared with linear decay for BiCG $(I-A)^{-1}v$ and theoretical bound (theorem and Taylor remainder)

Error bound for skew-Hermitian matrix



Error bound for skew-Hermitian matrix source



Lanczos method

- Second basis $W_m = (w_1, \dots, w_m)$ of Krylov subspace $K_m(A^*, w_1)$
- Biorthogonality condition $(v_i, w_j) = \delta_{ij}$, $D_m = W_m^* V_m$ block diagonal
- $H_m = D_m^{-1} W_m^* A V_m$ block triagonal, short recurrences
- $\mathcal{F}(H_m) \not\subset \mathcal{F}(A)$, oblique projection
- pseudospectrum $\Lambda_\varepsilon(A) = \{\lambda \in \mathbb{C} : ||(\lambda I - A)^{-1}|| \geq \varepsilon^{-1}\}$
- Condition number of eigenvectors matrix $\kappa_e(A) = ||X|| \cdot ||X^{-1}||$

Error bounds

- $\Lambda_e(A) \subset E, \Lambda_\gamma(A) \bigcup \Lambda_\gamma(H_m) \subset G$
- Error bound lemma with $M = \frac{1}{2}(1 + ||V_m|| \cdot ||D_m^{-1} W_m^*||) l(\partial E)/(\varepsilon\gamma)$
- A diagonalizable, spectrum $\Lambda(A) \subset E, \Lambda(H_m) \subset G$, $||(\lambda I - H_m)^{-1}|| \leq \gamma^{-1}$
- Error bound lemma with $M = 3\kappa_\varepsilon(A)(\delta^{-1} + ||V_m|| \cdot ||D_m^{-1} W_m^*|| \gamma^{-1})$, δ is distance between $\Lambda(A)$ and Γ

Examples

No graphs in paper, consider some examples to show bounds.
Last section about ODE schemes could be omitted, no graphs just scheme definition.